

Proof. 1. **Norm induced by the inner product.** By definition,

$$\|v\| = \sqrt{\langle v, v \rangle} \quad \text{for every } v \in V.$$

2. **Equivalence with the squared inequality.** Since norms are non-negative,

$$\|x + y\| \leq \|x\| + \|y\| \iff \|x + y\|^2 \leq (\|x\| + \|y\|)^2.$$

3. **Expansion of the left-hand side.**

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned}$$

4. **Expansion of the right-hand side.**

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2.$$

5. **Cauchy–Schwarz inequality.** For any inner-product space,

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle = \|x\|^2 \|y\|^2,$$

hence

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Because $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$, we obtain

$$\operatorname{Re} \langle x, y \rangle \leq \|x\| \|y\|.$$

6. **Insertion of the bound.** Substituting the previous estimate into the expansion of $\|x + y\|^2$ gives

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

7. **Taking square roots.** Both sides are non-negative, so

$$\|x + y\| \leq \|x\| + \|y\|.$$

8. **Equality condition.** Equality holds iff equality holds in the Cauchy–Schwarz step, i.e. iff x and y are linearly dependent and $\langle x, y \rangle$ is a non-negative real number. Thus there exists $\lambda \geq 0$ such that $y = \lambda x$, and for such vectors the inequality becomes an equality.

□